

Short-Term Wind Power Scenario Generation Based on Conditional Latent Diffusion Models

Xiaochong Dong^{ID}, Zhihang Mao^{ID}, Yingyun Sun^{ID}, and Xinzhi Xu^{ID}

Abstract—Quantifying short-term uncertainty in wind power plays a crucial role in power system decision-making. In recent years, the scenario generation community has conducted numerous studies employing generative models. Among these generative models, diffusion models have shown remarkable capabilities with excellent posterior representation. However, diffusion models are seldom used to quantify renewable energy uncertainty. To fill this research gap, this manuscript proposes a novel conditional latent diffusion model (CLDM) adapted for short-term scenario generation. CLDM decomposes the wind power scenario generation task into deterministic forecasting and forecast error scenario generation. The embedding network is used to regress deterministic forecasts, which reduces the denoising complexity of diffusion models. The denoising network generates forecast error scenarios in a latent space. Subsequently, the wind power scenarios are reconstructed by combining deterministic forecasts and forecast error scenarios. The case study compares with existing state-of-the-art methods, CLDM demonstrates superior evaluation metrics and enhances the denoising efficiency.

Index Terms—Diffusion model, generative model, latent space, probabilistic forecasting, scenario generation, short-term, wind power.

I. INTRODUCTION

GLOBALLY, 77.6 GW of new wind power capacity was connected to power grids in 2022, bringing total installed wind capacity to 906 GW, a year-on-year growth of 9%. The 1-TW of wind power is expected to be installed by 2023 [1]. With the rise of installed wind power capacity, operators are gradually paying attention to the accuracy of wind power forecasting. Especially for short-term (hours to days) wind power forecasting, which is inextricably linked with the unit commitment, power market trading, and economic dispatch of power systems [2], [3], [4]. Deterministic forecasts are commonly used to assess forecasting accuracy in power systems [5], [6]. In China, system

operators set strict requirements of short-term deterministic forecast accuracy for grid-connected wind farms, with penalties that break the forecast error trigger point [7].

Since the atmospheric system is chaotic, short-term wind power cannot be accurately forecasted. Therefore, it is necessary to model the uncertainty of wind power or forecast error. Considering the correlation between multiple periods, the T moments wind power or forecast error should be considered a stochastic process, which can be described by a T -dimensional joint probability distribution. The modeling of the joint probability distribution can be approached from two perspectives, the first modeling perspective is “*decoupling modeling*”. The joint probability distribution can be decoupled into multiple marginal probability distributions and a correlation structure of multivariate.

Decoupling modeling is commonly solved by regression models, which regress the specific point or function parameters of the marginal probability distributions and correlation structure. According to the expression form of marginal probability distributions, the major modeling forms include probability density function (PDF) and quantile function (QF). PDF models the probability density of marginal probability distributions. Parametric PDF assumes specific distributions for variables, such as Gaussian, Beta, t -location, Weibull, etc. [8], [9]. However, the parametric PDF is limited by these assumptions, and its generalization performance can be improved through mixture density estimation [10] or kernel density estimation [11]. On the other hand, the QF represents the inverse function of the cumulative density function (CDF). Non-parametric quantile regression is often used to estimate multiple quantiles for modeling the QF [12]. The quantile regression must prevent quantile crossover due to the monotonic increase of the QF [13]. While discrete quantiles are not constrained by parametric assumptions, discrete quantile regression cannot model marginal probability distributions continuously.

Marginal probability distributions cannot capture the correlation of multivariate. To obtain the joint probability distribution, correlation structure modeling is necessary. This typically involves methods such as time series models, Gaussian processes, Markov chains, and Copula functions. Time series models use linear regression functions to capture correlations between variables [14]. Gaussian processes use covariance to model correlations between Gaussian marginal distributions [15]. Markov chains provide non-parametric correlation modeling but are only applicable to discrete empirical marginal distributions

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[16]. Copula functions connect different types of marginal distributions through a joint CDF, but their effectiveness is limited by assumptions of the Copula function [17].

Modeling joint probability distributions using regression models is an analytical and highly interpretable approach that relies on strict regression scores. However, making optimization decisions based on the analytical joint probability distribution becomes challenging. Stochastic optimization requires a discrete scenario set to describe uncertainty [18]. When a T -dimensional time series is sampled from the joint probability distribution, it is referred to as a time-series scenario. By leveraging the law of large numbers, repeated sampling generates a finite number of discrete scenarios that approximate the joint probability distribution. Thus, the second modeling perspective is “*mapping modeling*”, which uses generative models to map discrete scenarios that conform to the joint probability distribution. The generative models do not rely on the regression of specific point or function parameters, nor decouple the joint probability distribution. Instead, a prior probability distribution is mapped to the joint probability distribution by deep neural networks [19].

The goal function of generative models is to maximize likelihood estimation. However, explicitly expressing the likelihood function of the joint probability distribution is challenging, leading to the development of various generative models to address this training problem. Generative adversarial networks (GAN) use the generator and discriminator in a zero-sum game, effectively bypassing the likelihood function. Under the Wasserstein distance loss function, GAN exhibits excellent performance [20]. However, the game training process introduces instability. Normalizing flows (NF) addresses the maximum likelihood function using the Jacobi matrix, but this approach limits the range of network structures and dimensions [21]. On the other hand, variational autoencoders (VAE) and diffusion models use the evidence lower bound (ELBO) to approximate the likelihood function, resulting in improved stability [22], [23]. However, the performance of the VAE is limited by the expressiveness of its variational posterior distribution. On the contrary, diffusion models address the issue of posterior expression through an explicit diffusion process, rendering it an excellent generative model for data synthesis [24]. Nevertheless, the multi-step denoising process increases the computational complexity of the diffusion models.

Since the introduction of generative models into scenario generation, there has been rapid development. Numerous researches have demonstrated the feasibility of using generative models for wind, photovoltaic, and load scenario generation [25]. To enhance the practicality of scenario generation, controllable generation has emerged as an important research branch. Chen et al. [26] incorporated class information to generate controllable features and seasonal scenarios. Dong et al. [27] devised continuously controllable features to generate scenarios with novel patterns. Kang et al. [28] integrated geographical information to generate spatio-temporal renewable energy scenarios. For short-term scenario generation, numerical weather prediction (NWP) or deterministic forecasts can be used as conditional values for the generative models [29], [30]. The crucial goal of short-term scenario generation is to ensure forecast accuracy, in

other words, to make the generated scenarios close to the measured power. This aspect also makes it different from traditional data synthesis [31].

Given the complexity of conditional information and generated scenarios, embedding networks and latent space have become indispensable components of generative models. Various neural networks can be used as embedding networks, including multilayer perceptron, convolutional neural networks, recurrent neural networks, and attention-based neural networks [32]. While embedding networks are typically integrated into generative models, this integration increases training complexity. By using an independent embedding network, the tasks of feature extraction and scenario mapping can be decomposed, thereby enhancing the controllability of scenario generation [33]. In the case of short-term scenario generation, the NWP data comprises numerous meteorological features, both relevant and irrelevant features [34]. Consequently, an independent embedding network is necessary to filter the meteorological features during short-term scenario generation.

Latent space modeling serves to simplify complex data using dimensionality reduction or data decomposition, thereby reducing the computational complexity and focusing on key feature mappings. However, existing latent space modeling approaches in scenario generation rely on decoupling the prior distribution, which is limited by feature entanglement [35]. In recent years, the VAE-based latent space models have been demonstrated in image synthesis [36] and super-resolution [37]. Nevertheless, the posterior representation of VAE may impact the accuracy of short-term scenario generation, which is undesirable. Consequently, it is necessary to design more appropriate latent space models to address this limitation.

In summary, there are several challenges to introducing diffusion models into short-term scenario generation. In this work, we propose a novel latent diffusion model adapted for short-term scenario generation, which makes the following key contributions:

- 1) The proposed CLDM uses NWP as a conditional value to realize short-term wind power scenario generation. CLDM decomposes the short-term scenario generation task into deterministic forecast regression and forecast error scenario generation. The diffusion process occurs within the latent space of forecast error.
- 2) The embedding network is proposed to regress deterministic forecasts and effectively filter out irrelevant NWP features. The adoption of a strict regression loss function ensures the accuracy of short-term scenario generation.
- 3) The denoising network maps the forecast error scenarios by the latent space, conditioned on the deterministic forecasts. This reduces the complexity of the denoising process and generates high-quality scenarios with fewer diffusion steps.

The rest of this manuscript is organized as follows. Section II provides the background of the diffusion models. Section III presents the network structure and algorithms of the CLDM. Section IV summarizes the case study. Section V presents the numerical results. Finally, Section VI concludes the manuscript.

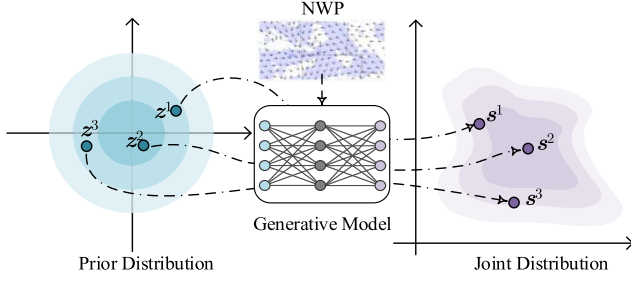


Fig. 1. Mapping relationships of generative models.

II. DIFFUSION MODELS

A. Problem Formulation

This section introduces the problem of short-term scenario generation using generative models. In practical operations, short-term forecasts need to be submitted one day in advance. For example, the New York Independent System Operator (NY-ISO) mandates day-ahead forecasts to be updated twice daily at 4 AM and 4 PM on the previous day [5]. Similarly, in China, system operators require day-ahead forecasts updated at 8 AM on the previous day [38].

Assume that the forecast lead time is t . When T periods NWP data $\mathbf{x}_{t:t+T}$ are used as covariate vectors, the T periods wind power $\mathbf{y}_{t:t+T}$ can be expressed by the conditional joint probability distribution $q(\mathbf{y}_{t:t+T}|\mathbf{x}_{t:t+T})$. Scenario $\mathbf{s}_{t:t+T}$ is a sample by $q(\mathbf{y}_{t:t+T}|\mathbf{x}_{t:t+T})$ as

$$\mathbf{s}_{t:t+T} \sim q(\mathbf{y}_{t:t+T}|\mathbf{x}_{t:t+T}) \quad (1)$$

When the $q(\mathbf{y}_{t:t+T}|\mathbf{x}_{t:t+T})$ is repeatedly sampled M times, a scenario set S is obtained as

$$S = \{\mathbf{s}^1, \mathbf{s}^2, \dots, \mathbf{s}^m, \dots, \mathbf{s}^M\} \quad (2)$$

The scenario $\mathbf{s}_{t:t+T}$ can be mapped from a prior distribution $p(\mathbf{z})$ to $q(\mathbf{y}_{t:t+T}|\mathbf{x}_{t:t+T})$ by a neural network as

$$\mathbf{s}_{t:t+T} = G(\mathbf{z}, \mathbf{x}_{t:t+T}, \theta) \sim q(\mathbf{y}_{t:t+T}|\mathbf{x}_{t:t+T}) \quad (3)$$

where G is the mapping function of generative models; θ is the learnable parameter of the neural network. For ease of notation, $\mathbf{x}$ and $\mathbf{y}$ are used instead of $\mathbf{x}_{t:t+T}$ and $\mathbf{y}_{t:t+T}$ in the following.

Therefore, generative models provide a continuous mapping of the joint probability distribution without relying on specific parameter assumptions. The mapping relationship of the generative model is shown in Fig. 1.

B. Diffusion and Denoising Process

The diffusion models consist of two components: a diffusion process and a denoising process, which are inversely related [39]. In the diffusion process, prior noise is iteratively added to the time series, causing it to converge towards a prior distribution. Conversely, the denoising process involves iteratively removing noise from the prior distribution to generate the time series. Fig. 2 illustrates the diffusion and denoising processes applied to the wind power time series, where the diffusion step is N . The

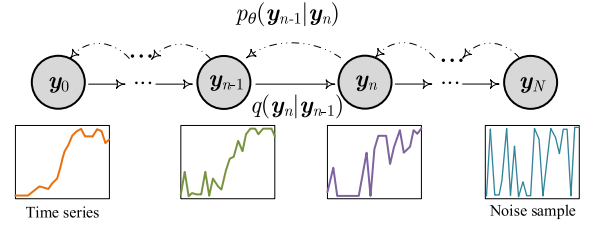


Fig. 2. Diffusion and denoising process of time series.

forward process is a diffusion process and the reverse process is a denoising process.

1) *Diffusion Process*: The time series sample is $\mathbf{y}_0 \sim p(\mathbf{y}_0)$, and the n -th diffusion time series sample is $\mathbf{y}_n$. Assuming that the prior distribution is a standard multivariate Gaussian distribution $\mathbf{y}_N \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The diffusion process from $\mathbf{y}_{n-1}$ to $\mathbf{y}_n$ is explicit as

$$\mathbf{y}_n = \sqrt{1 - \beta_n} \mathbf{y}_{n-1} + \beta_n \epsilon, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (4)$$

where $\beta \in (0, 1)$ is the scaling factor. The probability distribution of $\mathbf{y}_n$ as

$$q(\mathbf{y}_n|\mathbf{y}_{n-1}) = \mathcal{N}(\sqrt{1 - \beta_n} \mathbf{y}_{n-1}, \beta_n \mathbf{I}) \quad (5)$$

The $\mathbf{y}_n$ can be expressed as $\mathbf{y}_0$ over an n -order Markov process as

$$q(\mathbf{y}_n|\mathbf{y}_0) = \prod_n q(\mathbf{y}_n|\mathbf{y}_{n-1}) \quad (6)$$

Let $\alpha_n = 1 - \beta_n, \bar{\alpha}_n = \prod_n \alpha_n$. The relationship between $\mathbf{y}_n$ and $\mathbf{y}_0$ can be expanded as

$$\begin{aligned} \mathbf{y}_n &= \sqrt{\alpha_n} \mathbf{y}_{n-1} + \sqrt{1 - \alpha_n} \epsilon \\ &= \sqrt{\alpha_n} \left(\sqrt{\alpha_{n-1}} \mathbf{y}_{n-2} + \sqrt{1 - \alpha_{n-1}} \epsilon \right) + \sqrt{1 - \alpha_n} \epsilon \\ &= \sqrt{\alpha_n \alpha_{n-1} \alpha_{n-2} \dots \alpha_0} \mathbf{y}_0 + \sqrt{1 - \alpha_n \alpha_{n-1} \alpha_{n-2} \dots \alpha_0} \epsilon \\ &= \sqrt{\bar{\alpha}_n} \mathbf{y}_0 + \sqrt{1 - \bar{\alpha}_n} \epsilon \end{aligned} \quad (7)$$

The probability distribution $\mathbf{y}_n$ can be transformed as [39]

$$q(\mathbf{y}_n|\mathbf{y}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_n} \mathbf{y}_0, (1 - \bar{\alpha}_n) \mathbf{I}) \quad (8)$$

The diffusion process can be obtained directly using (8), eliminating the need for iterative solutions. The diffusion process resembles the mapping performed by the VAE encoder, which transforms the sample distribution into a posterior distribution. However, the key distinction lies in the explicit diffusion process of the diffusion model, whereas the VAE encoding relies on neural networks for its transformation.

2) *Denoising Process*: The denoising process corresponds to the mapping relationship in Fig. 1, where the generative sample distribution $p_\theta(\mathbf{y}_0)$ is denoised from the prior distribution $p(\mathbf{y}_N)$. The denoising process requires fitting a conditional probability distribution with a neural network $p_\theta(\mathbf{y}_{n-1}|\mathbf{y}_n)$. It can be expressed as an n -order Markov process as

$$p_\theta(\mathbf{y}_0) = p(\mathbf{y}_N) \prod_n p_\theta(\mathbf{y}_{n-1}|\mathbf{y}_n) \quad (9)$$

where $p_\theta(\mathbf{y}_{n-1}|\mathbf{y}_n)$ is a multivariate Gaussian distribution obtained by the learnable parameters θ .

The goal of the diffusion model is to achieve equality between the posterior distribution $p_\theta(\mathbf{y}_{n-1}|\mathbf{y}_n)$ and the prior distribution $q(\mathbf{y}_{n-1}|\mathbf{y}_n)$. However, directly accessing the prior distribution $q(\mathbf{y}_{n-1}|\mathbf{y}_n)$ is not straightforward. To address this, the prior distribution $q(\mathbf{y}_{n-1}|\mathbf{y}_n)$ is directly transformed by the Bayesian formula as

$$q(\mathbf{y}_{n-1}|\mathbf{y}_n, \mathbf{y}_0) = \frac{q(\mathbf{y}_n|\mathbf{y}_{n-1}, \mathbf{y}_0)q(\mathbf{y}_{n-1}|\mathbf{y}_0)}{q(\mathbf{y}_n|\mathbf{y}_0)} \quad (10)$$

where three terms can be transformed by (7) as

$$\begin{cases} q(\mathbf{y}_n|\mathbf{y}_{n-1}, \mathbf{y}_0) = q(\mathbf{y}_n|\mathbf{y}_{n-1}) = \sqrt{\alpha_n}\mathbf{y}_{n-1} + \sqrt{1-\alpha_n}\epsilon \in \\ q(\mathbf{y}_n|\mathbf{y}_0) = q(\mathbf{y}_n|\mathbf{y}_0) = \sqrt{\alpha_n}\mathbf{y}_0 + \sqrt{1-\alpha_n}\epsilon \in \\ q(\mathbf{y}_{n-1}|\mathbf{y}_0) = q(\mathbf{y}_{n-1}|\mathbf{y}_0) = \sqrt{\alpha_{n-1}}\mathbf{y}_0 + \sqrt{1-\alpha_{n-1}}\epsilon \in \end{cases} \quad (11)$$

Substituting (11) into (10) gives

$$q(\mathbf{y}_{n-1}|\mathbf{y}_n, \mathbf{y}_0) = \mathcal{N}\left(\frac{1}{\sqrt{\alpha_n}}\left(\mathbf{y}_n - \frac{\beta_n}{\sqrt{1-\alpha_n}}\epsilon\right), \bar{\beta}_n\mathbf{I}\right) \quad (12)$$

where $\bar{\beta}_n = \beta_n(1-\alpha_{n-1})/1-\alpha_n$.

From (12), it is evident that the variance is solely dependent on the hyperparameters β . Consequently, the denoising network needs to fit a noise sample ϵ_θ . As a result, the denoising process of the diffusion model is represented as

$$\mathbf{y}_{n-1} = \frac{1}{\sqrt{\alpha_n}}\left(\mathbf{y}_n - \frac{\beta_n}{\sqrt{1-\alpha_n}}\epsilon_\theta\right) + \bar{\beta}_n\epsilon \quad (13)$$

The posterior distribution $p_\theta(\mathbf{y}_{n-1}|\mathbf{y}_n)$ is represented as:

$$p_\theta(\mathbf{y}_{n-1}|\mathbf{y}_n) = \mathcal{N}\left(\frac{1}{\sqrt{\alpha_n}}\left(\mathbf{y}_n - \frac{\beta_n}{\sqrt{1-\alpha_n}}\epsilon_\theta\right), \bar{\beta}_n\mathbf{I}\right) \quad (14)$$

3) *Prior and Posterior Distributions*: In this section, the prior distribution assumption is analyzed to demonstrate the validity of diffusion models. Through variational inference, the likelihood function can be transformed into the ELBO, denoted as

$$\begin{aligned} \log p_\theta(\mathbf{y}_0) \\ \geq \mathbb{E}_{q(\mathbf{y}_N|\mathbf{y}_0)}[\log p_\theta(\mathbf{y}_0|\mathbf{y}_N) + \log p(\mathbf{y}_N) - \log q(\mathbf{y}_N|\mathbf{y}_0)] \end{aligned} \quad (15)$$

The objective of the variational model is to optimize the ELBO with respect to the sample distribution. However, due to the assumption of the prior distribution $p(\mathbf{y}_N)$, training generative models is required for both the generative sample distribution $p_\theta(\mathbf{y}_0|\mathbf{y}_N)$ and the posterior distribution $q(\mathbf{y}_N|\mathbf{y}_0)$. In the VAE, the posterior distribution obtained using an encoder may not align with the desired prior distribution [40].

If the prior distribution $p(\mathbf{y}_N)$ is stationary, the Markov process can map the sample distribution $q(\mathbf{y}_0)$ to the posterior distribution $q(\mathbf{y}_N|\mathbf{y}_0)$. The probability path is obtained as (7). It is important to note that $\alpha_n \in (0, 1)$, which implies that $\lim_{n \rightarrow N} \bar{\alpha}_n = 0$. As the diffusion process progresses through the N -th order, the posterior distribution naturally converges to

the prior distribution as

$$\lim_{\bar{\alpha}_n \rightarrow 0} q(\mathbf{y}_N|\mathbf{y}_0) = p(\mathbf{y}_N) \quad (16)$$

This natural diffusion process does not require specific parametric assumptions. Various prior distributions can be selected for the diffusion process. Increasing the degrees of freedom in the prior distributions can expedite the convergence of the diffusion process. However, the choice of prior distribution does not impact the convergence of the posterior distribution. On the other hand, using complex prior distributions may require costly computation [41]. Under comprehensive consideration, the standard multivariate Gaussian distribution is selected as the prior distribution. It guarantees convergence of the posterior distribution and does not compromise the accuracy of the probability mapping.

III. CONDITIONAL LATENT DIFFUSION MODELS

A. Generative Modeling of Latent Space

The traditional diffusion models obtain the generative sample distribution $p_\theta(\mathbf{y}_0|\mathbf{y}_N)$ by iterative denoising of the prior distribution $p(\mathbf{y}_N)$. In short-term scenario generation, the diffusion model is mapped to conditional distribution $p(\mathbf{y}_0|\mathbf{x})$. The NWP data $\mathbf{x}$ includes various meteorological factors and plays a crucial role. Among these factors, wind speed exhibits a strong correlation with wind power, while other meteorological factors may not exhibit such a clear relationship. When using NWP data as conditions for the diffusion network, the diffusion network not only fits noise ϵ_θ but also filters the NWP features. Consequently, this leads to an inevitable increase in computational complexity.

In the realm of image synthesis, the most successful generative models use a mapping from a low-dimensional latent space to generate images [42]. This approach not only reduces computational complexity but also yields higher-quality mappings. Drawing inspiration from this methodology, we propose a latent space mapping-based method for short-term scenario generation. However, unlike image synthesis, the method proposed does not involve mapping to a low-dimensional latent space. Instead, the short-term wind power scenario is decomposed into two components: deterministic forecasts $\hat{\mathbf{y}}$ and forecast errors $\Delta\mathbf{y}_0$. Deterministic forecasts $\hat{\mathbf{y}}$ are obtained from an independent embedding network. The diffusion process maps the forecast error distribution $p(\Delta\mathbf{y}_0)$, which is a novel latent space, as shown in Fig. 3.

The CLDM consists of three modules. The embedding module regresses wind power deterministic forecasts $\hat{\mathbf{y}}$ from NWP data $\mathbf{x}$. The $\hat{\mathbf{y}}$ is used as the conditional value for the denoising process and latent space conversion.

The latent space diffusion module comprises two processes: diffusion and denoising. In the diffusion process, the forecast error $\Delta\mathbf{y}_0$ is transformed into posterior sample $\Delta\mathbf{y}_N \sim q(\Delta\mathbf{y}_N|\Delta\mathbf{y}_0)$. In the denoising process, after N recurrent denoising steps, the prior sample $\Delta\mathbf{y}_N \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ maps to the generative sample $\Delta\mathbf{y}_0 \sim p_\theta(\Delta\mathbf{y}_0|\Delta\mathbf{y}_N)$.

The conversion module is responsible for converting wind power time series and forecast errors. During the training stage,

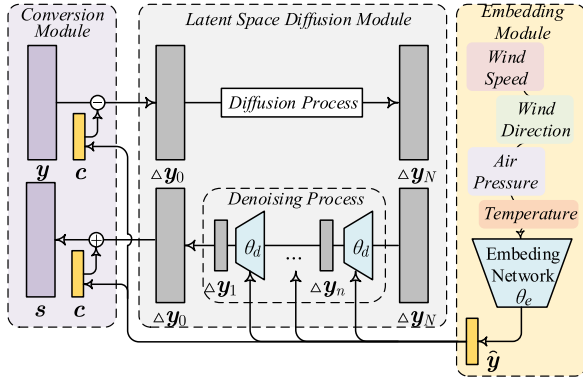


Fig. 3. Structure of CLDM.

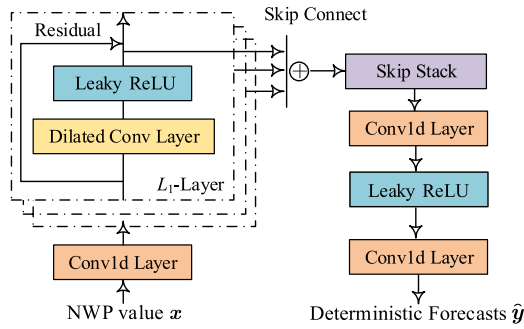


Fig. 4. Structure of embedding network.

the forecast error $\Delta \mathbf{y}_0$ is obtained by subtracting the deterministic forecasts $\hat{\mathbf{y}}$ from the measured wind power $\mathbf{y}$. During the testing stage, the wind power scenario $\mathbf{s}$ is obtained by adding the deterministic forecasts $\hat{\mathbf{y}}$ to forecast error scenarios $\Delta \mathbf{y}_0$ as

$$\mathbf{s} = \hat{\mathbf{y}} + \Delta \mathbf{y}_0 = f_d(\mathbf{z}, \mathbf{x}, \theta_d) + f_e(\mathbf{x}, \theta_e) \quad (17)$$

where θ_d and θ_e are the learnable parameters of the denoising network and embedding network, respectively; $f_d(\sim)$ and $f_e(\sim)$ are the mapping functions of denoising network and embedding network, respectively.

In the scenario generation task, the embedding module plays a crucial role in filtering NWP data. The deterministic forecasts serve as conditions for the denoising network. By performing denoising in the latent space, the computational complexity of the denoising network is effectively reduced. Next, the structure of the embedding and denoising networks is described in detail.

1) *Embedding Network*: The embedding network consists of L_1 -layer dilated convolutional layers for extracting NWP features, as shown in Fig. 4. The input and output layers of the embedding network use 1-dimensional convolution to adjust the number of channels. The embedding network uses LeakyReLU as the activation function. By multi-layer dilated convolutional layers, the perceptual field of the network for time series can be increased. Residual connections are incorporated to mitigate network degradation, while skip connections are used to merge features across multi-timescale.

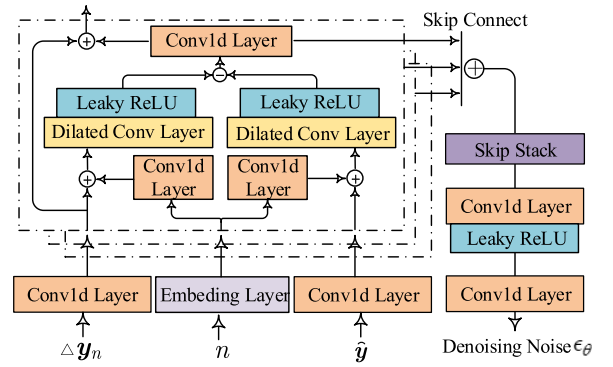


Fig. 5. Structure of denoising network.

2) *Denoising Network*: The structure of the denoising network is shown in Fig. 5. The input to the denoising network consists of denoising samples $\Delta \mathbf{y}_n$, step n , and deterministic forecasts $\hat{\mathbf{y}}$, while the output is a denoising noise ϵ_θ .

The wind power forecast error includes a positive correlation between adjacent moments. The denoising process maps the uncorrelation prior sample $\Delta \mathbf{y}_N \sim \mathcal{N}(\mathbf{0}, \mathbf{1})$ to a positive correlation error scenario $\Delta \mathbf{y}_0$. As indicated by (18), the denoising noise ϵ_θ is the sole factor influencing the correlation during the denoising process. Therefore, the denoising process must obtain a negative correlation noise ϵ_θ to gradually map the uncorrelation sample $\Delta \mathbf{y}_N$ to the positive correlation error scenario $\Delta \mathbf{y}_0$ as

$$\Delta \mathbf{y}_n = \frac{1}{\sqrt{\alpha_n}} \left(\Delta \mathbf{y}_{n-1} - \frac{\beta_n}{\sqrt{1 - \alpha_n}} \epsilon_\theta \right) + \bar{\beta}_n \in \quad (18)$$

The embedding layer maps the discrete step n to a continuous vector space. Subsequently, the embedding step element-wise adds to the denoising sample and deterministic forecasts through 1-dimensional convolution. An element-wise subtraction is used to obtain the negative correlation feature. Similar to the embedding network, L_2 -layer dilated convolutional layers, residual connections, and skip connections are used to merge multi-timescale features.

B. Scenario Generation Algorithms of CLDM

The training of the CLDM comprises two primary components: the embedding network and the denoising network. The regression loss function for the embedding network can be chosen from mean absolute error (MAE) or root mean square error (RSME). The loss function can be chosen according to the forecasting demand. The MAE loss function is suitable for obtaining median forecasts, while the RMSE loss function is suitable for obtaining expected forecasts [43]. In this manuscript, the MAE loss function is used for training the embedding network.

For the diffusion network, the Kullback-Leibler (KL) divergence of $p_\theta(\Delta \mathbf{y}_{n-1} | \Delta \mathbf{y}_n)$ and $q(\Delta \mathbf{y}_n | \Delta \mathbf{y}_{n-1})$ is used as the loss function, which is equivalent to the ELBO of the likelihood function. Based on the KL divergence of the multivariate

Gaussian distribution, the loss function can be transformed as

$$\begin{aligned} l_{\text{ELBO}} &= KL(p_{\theta}(\Delta \mathbf{y}_{n-1} | \Delta \mathbf{y}_n) || q(\Delta \mathbf{y}_{n-1} | \Delta \mathbf{y}_n)) \\ &= \mathbb{E}_{\Delta \mathbf{y}_0, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \left[\frac{\beta_n^2}{2\alpha_n(1 - \bar{\alpha}_n) ||\bar{\beta}_n \mathbf{I}||^2} ||\epsilon - \epsilon_{\theta}||^2 \right] \end{aligned} \quad (19)$$

where $|| \cdot ||$ is the Euclidean norm. After removing the constant weights, (19) is transformed as [44]

$$l_{\text{ELBO}} = \mathbb{E}_{\Delta \mathbf{y}_0, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} ||\epsilon - \epsilon_{\theta}(\Delta \mathbf{y}_n, n, \hat{\mathbf{y}})||^2 \quad (20)$$

The pseudo-code for training and scenario generation of CLDM is summarized in Algorithm 1 and Algorithm 2.

IV. CASE STUDY

A. Data Source Description and Experiment Settings

The dataset was derived from Zone 1 of the Global Energy Forecasting Competition 2014 [45]. The cases considered day-ahead wind power scenario generation, with the model using 24-hour ahead NWP data. The raw wind power data is represented in per-unit value, while the NWP data includes zonal and meridional wind components at two altitudes (10 and 100 m). The training set consisted of the 2012 dataset, while the 2013 dataset served as the test set for evaluating forecasting performance. To create a validation set, 20% of the samples were randomly selected from the training set. The case was implemented on a computer equipped with NVIDIA TITAN V graphics card and an Intel Core i9-7900X 3.30 GHz processor. The experimental model was developed using the PyTorch 1.8.0 backend. For all deep learning models, the Adam optimization algorithm was used during training. Table I provides a summary of the hyperparameter settings used in the CLDM training process.

B. Benchmark Models

The case study compares the five existing state-of-the-art scenario generation models with each other. NWP data is used as conditional values for short-term scenario generation.

DeepAR [46] is used as the benchmark regression model. The expected value and variance of a Gaussian distribution as its output. On this basis, the Gaussian copula function is used to model the correlation structure of multivariate. The loss function is the likelihood function.

WGAN [25] contains a discriminator and a generator network. The generator network maps samples from the prior samples to wind power scenarios. The loss function is the Wasserstein distance based on the gradient penalty.

VAE [22] contains an encoder and a decoder network. The decoder network maps samples from the prior distribution to wind power scenarios. The loss function is the ELBO.

NF [47] uses an affine coupling layer derived from real-valued non-volume preserving to achieve network reversibility. The inverse mapping of the NF network generates wind power scenarios. The loss function is the likelihood function.

Algorithm 1: Training Procedure of the CLDM.

Input: The NWP data $\mathbf{x}$; the measured wind power $\mathbf{y}$; the network layer $\{L_1, L_2\}$; the diffusion step N ; the scaling factor β ; the learning rate $\{lr_1, lr_2\}$; the batch size b ; the epoch number $\{h_1, h_2\}$.

Embedding network training

while θ_e **not** converged **do**

for each epoch in range(h_1) **do**

Forward propagation

 embed mapping: $\hat{\mathbf{y}} \leftarrow f_e(\mathbf{x}, \theta_e)$

 loss: $l_{\text{MAE}} \leftarrow \mathbb{E} ||\mathbf{y} - \hat{\mathbf{y}}||$

Back propagation

$\theta_e \leftarrow \text{Adam}(\nabla l_{\text{MAE}}, lr_1)$

end for

end while

Denoising network training

$\alpha_n \leftarrow 1 - \beta_n, \bar{\alpha}_n \leftarrow \prod_n \alpha_n$

while θ_d **not** converged **do**

for each epoch in range(h_2) **do**

Forward propagation

 latent space mapping: $\Delta \mathbf{y}_0 \leftarrow \mathbf{y} - \hat{\mathbf{y}}$

 random diffusion step: $n \leftarrow \text{uniform}\{1, \dots, N\}$

 diffusion sample: $\Delta \mathbf{y}_n \leftarrow \sqrt{\bar{\alpha}_n} \Delta \mathbf{y}_0 + \sqrt{1 - \bar{\alpha}_n} \epsilon$

 noise generation: $\epsilon_{\theta} \leftarrow f_d(\Delta \mathbf{y}_n, n, \hat{\mathbf{y}}, \theta_d)$

 loss: $l_{\text{ELBO}} \leftarrow \mathbb{E} ||\epsilon - \epsilon_{\theta}||^2$

Back propagation

$\theta_d \leftarrow \text{Adam}(\nabla l_{\text{ELBO}}, lr_2)$

end for

end while

return optimized parameters θ_e^*, θ_d^*

Algorithm 2: Scenario Generation of the CLDM.

Input: The NWP data $\mathbf{x}$; the number of scenarios M ; the scaling factor β ; the optimized parameters θ_e^*, θ_d^* .

Scenario generation

$\alpha_n \leftarrow 1 - \beta_n, \bar{\alpha}_n \leftarrow \prod_n \alpha_n$

$\bar{\beta}_n = \beta_n(1 - \bar{\alpha}_{n-1})/1 - \bar{\alpha}_n$

embedding NWP data: $\hat{\mathbf{y}} \leftarrow f_e(\mathbf{x}, \theta_e^*)$

for $m = 1, \dots, M$ **do**

$\Delta \mathbf{y}_N \leftarrow \mathcal{N}(\mathbf{0}, \mathbf{I})$

for $n = N, \dots, 1$ **do**

 noise generation: $\epsilon_{\theta} \leftarrow f_d(\Delta \mathbf{y}_n, n, \hat{\mathbf{y}}, \theta_d^*)$

 denoising: $\Delta \mathbf{y}_{n-1} \leftarrow \frac{1}{\sqrt{\alpha_n}}(\Delta \mathbf{y}_n - \frac{\bar{\beta}_n}{\sqrt{1 - \bar{\alpha}_n}} \epsilon_{\theta}) + \bar{\beta}_n \epsilon$

end for

 scenario generation: $\mathbf{s}^m \leftarrow \Delta \mathbf{y}_0 + \hat{\mathbf{y}}$

end for

return scenario set $S = \{\mathbf{s}^1, \mathbf{s}^2, \dots, \mathbf{s}^m, \dots, \mathbf{s}^M\}$

DDPM [39] is used as a benchmark for diffusion models. To ensure comparability, both DDPM and CLDM share the same denoising network structure. The difference is that DDPM does not include the embedding network or latent space mapping.

TABLE I
HYPERPARAMETER SETTINGS

Hyperparameter	Definition	Value
(β_0, β_N)	Scaling factor	$(10^{-4}, 0.05)$
N	Diffusion step	50
$\{lr_1, lr_2\}$	Learning rate	$\{10^{-3}, 10^{-3}\}$
$\{L_1, L_2\}$	Network layer	$\{3, 3\}$
b	Batch size	50
$\{h_1, h_2\}$	Epoch	$\{500, 500\}$
M	Number of scenarios	200

C. Evaluation Metrics

Multiple evaluation metrics are used to evaluate scenario generation from three perspectives: deterministic forecasting, probabilistic forecasting, and scenario generation. To prevent randomness, the case was repeated ten times to calculate average metrics. All metrics are averaged over the test samples.

1) *Deterministic Forecasting*: MAE and RMSE are used as evaluation metrics for deterministic forecasts which measure the difference between special points of the scenario set and the measured power. For the four benchmark generative models, the expected value of the scenario set $\bar{\mathbf{y}} = \mathbb{E}(\mathbf{s})$ is chosen as deterministic forecasts.

$$\text{MAE} = \frac{1}{T} \sum_{i=t+1}^{t+T} |\mathbf{y}_i - \bar{\mathbf{y}}_i| \quad (21)$$

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{i=t+1}^{t+T} (\mathbf{y}_i - \bar{\mathbf{y}}_i)^2} \quad (22)$$

2) *Probabilistic Forecasting*: The pinball score (PS) and continuous ranked probability score (CRPS) are used as evaluation metrics for marginal distribution probabilistic forecasting [48].

$$\text{PS}_u = \frac{1}{T} \sum_{i=t+1}^{t+T} \phi_i^u, \phi_i^u = \begin{cases} u(\mathbf{y}_i - \hat{\mathbf{y}}_i^u), & \mathbf{y}_i \geq \hat{\mathbf{y}}_i^u \\ (u-1)(\mathbf{y}_i - \hat{\mathbf{y}}_i^u), & \mathbf{y}_i < \hat{\mathbf{y}}_i^u \end{cases} \quad (23)$$

$$\text{CRPS} = \frac{1}{T} \sum_{i=t+1}^{t+T} \left(\frac{1}{M} \sum_{m=1}^M |\mathbf{s}_i^m - \mathbf{y}_i| - \frac{1}{2M^2} \sum_{n,m=1}^M |\mathbf{s}_i^m - \mathbf{s}_i^n| \right) \quad (24)$$

where u is the CDF value, which sets $u = 0.05 \sim 0.95$ with 0.05 interval; $\hat{\mathbf{y}}_i^u$ is the u -quantile forecasts; ϕ_i^u is an indicative function penalizing underestimation by u and overestimation by $u-1$.

Average coverage error (ACE) and predict interval average width (PIAW) are used to analyze the reliability and sharpness performance.

$$\begin{aligned} \text{ACE}_\tau &= \left(\frac{1}{T} \sum_{i=t+1}^{t+T} \chi_i^\tau - \tau \right) \times 100\%, \chi_i^\tau \\ &= \begin{cases} 1, & \mathbf{y}_i \in [\mathbf{s}_{i,\tau}^{\text{low}}, \mathbf{s}_{i,\tau}^{\text{upp}}] \\ 0, & \mathbf{y}_i \notin [\mathbf{s}_{i,\tau}^{\text{low}}, \mathbf{s}_{i,\tau}^{\text{upp}}] \end{cases} \end{aligned} \quad (25)$$

$$\text{PIAW}_\tau = \frac{1}{T} \sum_{i=t+1}^{t+T} (\mathbf{s}_{i,\tau}^{\text{upp}} - \mathbf{s}_{i,\tau}^{\text{low}}) \quad (26)$$

where τ is the confidence level; χ_i^τ is an indicative function, which identifies the measured values within the upper and lower bounds; $\mathbf{s}_{i,\tau}^{\text{low}}, \mathbf{s}_{i,\tau}^{\text{upp}}$ is the low and upper bound of the scenario set at τ confidence level.

3) *Scenario Generation*: The energy score (ES) and variogram score (VS) are used as evaluation metrics for scenario generation [49]. These metrics assess the correlation within variables and provide a quantitative measure of the quality of the generated scenarios.

$$\text{ES} = \frac{1}{M} \sum_{m=1}^M \|\mathbf{s}^m - \mathbf{y}\| - \frac{1}{2M^2} \sum_{n,m=1}^M \|\mathbf{s}^m - \mathbf{s}^n\| \quad (27)$$

$$\text{VS} = \sum_{i,j=t+1}^{t+T} \left(|\mathbf{y}_i - \mathbf{y}_j|^{1/2} - \frac{1}{M} \sum_{m=1}^M |\mathbf{s}_i^m - \mathbf{s}_j^m|^{1/2} \right)^2 \quad (28)$$

V. RESULTS AND DISCUSSION

A. Scenario Generation Results

This section presents a comparison of the modeling accuracy between benchmark models and CLDM in the short-term wind power scenario generation. The results for the six metrics are shown in Table II. Fig. 6 illustrates the results of scenario generation for the first three consecutive days in the test dataset. Furthermore, Fig. 7 provides a comparison of reliability and sharpness performance from 10% to 90% confidence level.

As shown in Table II, CLDM achieves the lowest values across all evaluation metrics, demonstrating its superior performance. Among the generative models, DDPM also outperforms WGAN, VAE, and NF.

In deterministic forecasting, CLDM demonstrates a 7.8% improvement in MAE and a 4.2% improvement in RMSE compared to DDPM. However, it is important to note that the DeepAR metrics offer a significant advantage over traditional generative models. This advantage arises from the fact that the traditional generative models do not rely on proper scoring rules during training. In contrast, the embedding network of CLDM adheres to a proper regression function, ensuring accurate deterministic forecasts. Therefore, integrating regression embedding into generative models is necessary.

In probabilistic forecasting, CLDM demonstrates a 5.0% improvement in CRPS and a 5.1% improvement in PS compared to DDPM. Moreover, for scenario generation, ES and VS exhibit respective improvements of 4.1% and 3.8% compared to DDPM. Considering that DDPM and CLDM share the same denoising network structure, the results indicate that accurate deterministic forecasts indirectly enhance the modeling accuracy in short-term scenarios.

In Fig. 6, it is evident that all six models successfully capture the short-term trend of wind power. Fig. 6(c) illustrates that VAE generates more centralized short-term scenarios, which are difficult to include the measured wind power. Fig. 7 measures

TABLE II
EVALUATION METRICS RESULTS

Model	MAE	RMSE	CRPS	PS	ES	VS
DeepAR	0.1241	0.1520	0.0874	0.0458	0.5325	16.2758
WGAN	0.1360	0.1667	0.0994	0.0519	0.6010	17.6408
VAE	0.1343	0.1626	0.1049	0.0541	0.6265	19.5108
NF	0.1371	0.1648	0.0953	0.0499	0.5757	17.4206
DDPM	0.1281	0.1547	0.0902	0.0473	0.5472	16.6070
CLDM	0.1180	0.1482	0.0857	0.0449	0.5248	15.9738

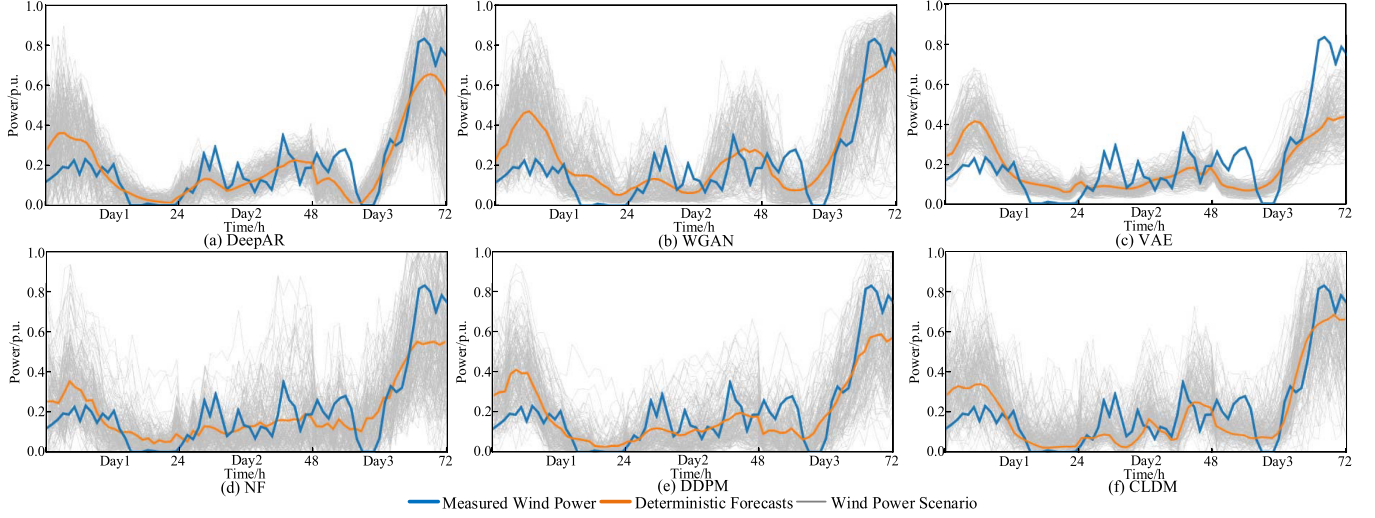


Fig. 6. Day-ahead wind power scenario generation results.

the reliability and sharpness performance of the models in uncertainty modeling. Specifically, Fig. 7(a) shows that DeepAR, NF, and CLDM have ACE metrics closer to zero, indicating their higher reliability compared to GAN and VAE.

When considering Fig. 7(b), it becomes apparent that VAE, GAN, and DDPM have lower PIAW values. However, lower ACE metrics suggest a lack of reliability. Among DeepAR, NF, and CLDM, which excel in ACE metrics, CLDM exhibits the smallest PIAW value. This implies that CLDM demonstrates superior reliability and sharpness simultaneously. In terms of comprehensive performance, CLDM demonstrates a high level of forecast accuracy while also maintaining reliability and sharpness.

B. Denoising Process Analysis

In this section, a comprehensive analysis of the denoising process of CLDM is provided. The analysis focuses on three aspects: the scenario generation process, the forecast error denoising process, and the effect of the diffusion step. To begin, the third day of the test set is selected as the case for testing. Ten random noise samples $\Delta \mathbf{y}_N$ are chosen for denoising. Here $d = N - n$ is defined as the denoising step. The diffusion model was then observed over four denoising steps ($d = 0, 30, 40, 50$) to demonstrate how it maps the noise samples into wind power scenarios. The results are presented in Fig. 8.

Fig. 8(a) shows the scenarios without denoising. Even after stacking the deterministic forecasts, it remains a chaotic scenario

that fails to capture the trend and uncertainty of wind power generation. In Fig. 8(b), after 30-step denoising, the scenarios exhibit reduced fluctuations compared to Fig. 8(a). Fig. 8(c) demonstrates a clear trend in wind power, although the fluctuations are still pronounced. Finally, in Fig. 8(d), after 50-step denoising, the scenarios display a high correlation in adjacent moments and exhibit the same power trend.

Next, we analyze the fitting process of the forecast error scenario using four denoising steps ($d = 0, 30, 40, 50$). This analysis provides a more direct assessment of the role of the denoising network. Fig. 9 displays scatter plots illustrating the difference between the measured error $\Delta \mathbf{y}$ and the forecast error scenarios $\Delta \mathbf{y}_n$.

The interval of wind power forecast error is $[\hat{\mathbf{y}} - 1, \hat{\mathbf{y}} + 0]$. The negative forecast error indicates an underestimation, while the positive error indicates an overestimation. The forecast error scenarios in Fig. 9(a) deviate from the measured errors, with many error scenarios located on the error boundary. After 30-step denoising in Fig. 9(b), the forecast error scenarios become denser in the low power interval. Subsequently, after 40-step denoising in Fig. 9(c), the forecast error scenario gradually decreases on the error boundary. Finally, after 50-step denoising in Fig. 9(d), the forecast error scenarios closely match the measured errors. It is observed that the wind power forecast error is smaller in the low power interval, while it is larger in the middle-high interval.

Finally, we discuss the effect of diffusion step N on the diffusion models. The computational efficiency of diffusion

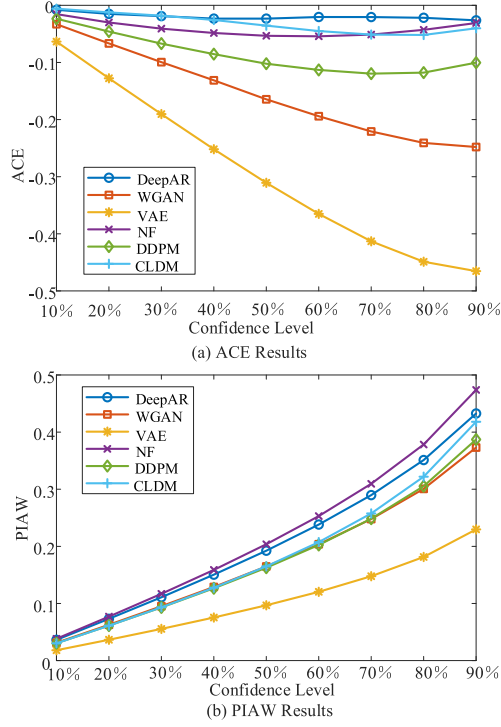
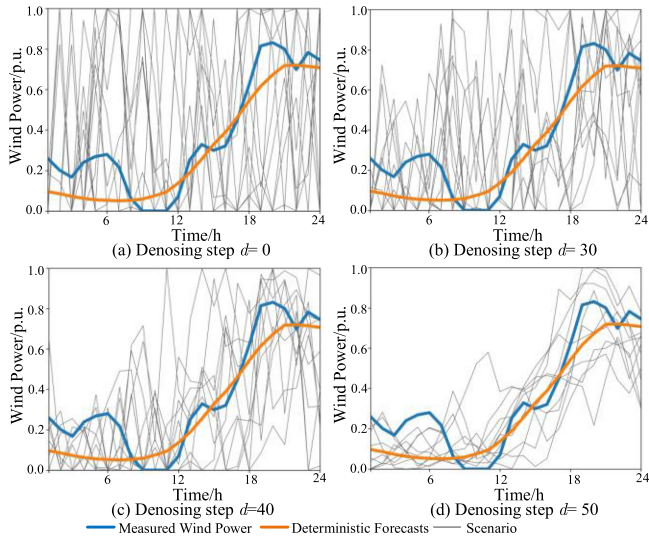


Fig. 7. Results of ACE and PIAW.

Fig. 8. Denoising process of wind power scenario generation. The scenarios are obtained by adding denoising samples $\Delta \mathbf{y}_n$ and deterministic forecasts $\hat{\mathbf{y}}$.

models is influenced by the diffusion step, particularly during the testing process where recurrent denoising is time-consuming. To enhance the computational efficiency of diffusion models, one straightforward approach is to reduce the diffusion step N . The case analysis assesses the accuracy of CLDM and DDPM using different diffusion steps ($N = 40, 30, 20, 10$). The evaluation metrics are compared with those in Table II ($N = 50$) by percent change, and the results are shown in Fig. 10.

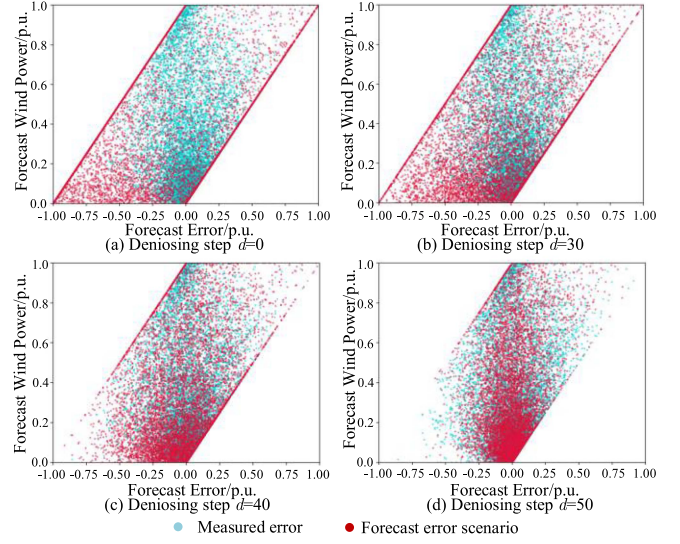


Fig. 9. Denoising process of forecast error. Discrete points contain full test set data.

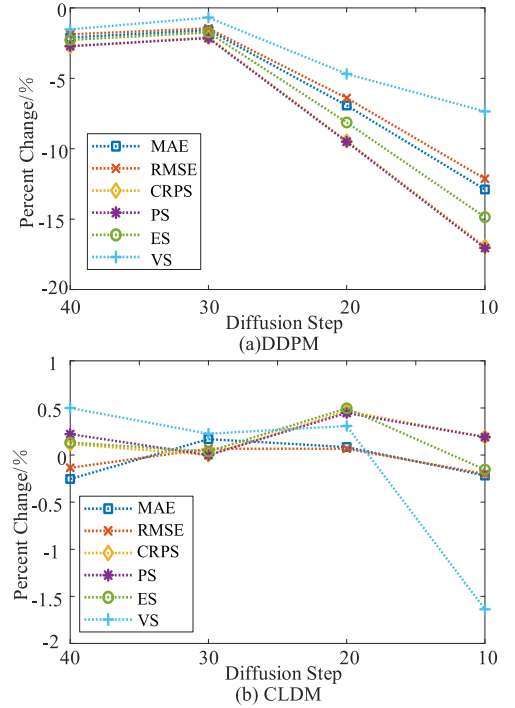


Fig. 10. Percent change of diffusion step for scenario generation.

The evaluation metrics of DDPM exhibit a noticeable decrease as the diffusion step decreases. When $N = 20$, the metrics decreased by 5~10%. Similarly, when $N = 10$, the metrics decreased by 7~15%. In contrast, CLDM demonstrates greater stability, with the percent change of each metric fluctuating within 0.5% as the diffusion step decreases to $N = 20$. When $N = 10$, the VS metric decreases by 1.5%, while the other metrics remain within a 0.5% percent change.

TABLE III
RESULTS OF STOCHASTIC UNIT COMMITMENT

Method	Total Cost ($\times 10^7$ \$)	Penalty Cost ($\times 10^7$ \$)	Load Shedding (MWh)	Wind Curtailment (MWh)
DeepAR	3.38	0.16	48.66	1389.59
WGAN	4.81	1.61	1513.25	1285.51
VAE	3.54	0.31	218.66	1052.69
NF	3.60	0.39	311.25	982.49
DDPM	4.64	1.44	1343.25	1257.93
CLDM	3.32	0.11	48.66	814.41

In CLDM, the metrics of MAE and RMSE are unaffected by the diffusion step N , as the accuracy is ensured by the embedding network. The diffusion step N primarily impacts the forecast error scenario. After regressing the deterministic forecasts, the latent space denoising can be performed with fewer diffusion steps. In DDPM, the denoising network is responsible for multiple tasks, including NWP feature filtering and sample distribution fitting. When the diffusion step is limited, it becomes challenging to maintain the accuracy of scenario generation.

C. Stochastic Unit Commitment

To further demonstrate the impact of scenario generation modeling on system operating costs, a stochastic unit commitment case study is simulated using the IEEE RTS-24 system [50]. The improved test system consists of ten thermal power units and three wind farms, with a thermal capacity of 2800 MW and a wind power capacity of 1200 MW.

The day-ahead unit commitment problem consists of two stages. In the first stage, the day-ahead scenarios are used as input to determine the on/off state of the thermal units. In the second stage, after the day-ahead unit commitment decisions are established, real-time economic dispatch is performed to calculate the dispatch cost. To account for the diverse nature of wind power, we generate day-ahead scenarios by using NWP data from Jan. 1st to Jan. 7th, 2013, and then simulate the day-ahead unit commitment for these seven days. The real-time economic dispatch costs are used to compare the impact of scenario generation results on system operation. In order to increase the efficiency of the computation, the K -medoids algorithm is used for scenario reduction with $K = 5$. The economic dispatch costs for these seven days are presented in Table III. The penalty for load shedding is set at 1000 \$/MWh, and the penalty for wind abandonment is set at 80 \$/MWh.

From Table III, it is evident that the CLDM exhibits the lowest dispatch cost, load shedding, and wind curtailment. In the test week, the system supplied a total of 270851 MWh of power. CLDM and DeepAR have the lowest load shedding, amounting to 48.66 MWh. The wind power generation reached a total of 53760 MWh. The scenarios generated by CLDM enabled the system to achieve the highest wind power utilization of 98.4%.

Two representative cases highlight the impact of inaccurate scenario generation on system operation. On Jan. 6th, from 09:00 to 18:00, both the WGAN and DDPM models exhibited a significant overestimation of wind power, leading to load shedding due to insufficient thermal reserve capacity. Similarly, between 1:00

and 7:00 on the same day, all models demonstrated a tendency to underestimate the wind power, resulting in wind curtailment due to the minimum output constraints of thermal units. Therefore, an accurate scenario generation method is crucial for achieving more economical short-term dispatch operations.

VI. CONCLUSION

In this study, traditional diffusion models have been extended to CLDM for short-term wind power scenario generation. CLDM divides the task of short-term scenario generation into two steps: deterministic forecast regression and forecast error scenario generation. The embedding network is used to regress highly accurate deterministic forecasts, while the denoising network maps forecast error scenarios in a novel latent space.

CLDM offers several advantages compared to existing methods. Firstly, it integrates regression models and generative models, combining the strengths of both approaches. The embedding network filters NWP features and uses a strict regression loss function to ensure the accuracy of scenario generation. Secondly, the denoising network enables continuous mapping of non-parametric forecast error scenarios. The use of latent space mapping reduces computational complexity. The case study demonstrates that CLDM exhibits higher accuracy than benchmark models and superior computational efficiency compared to DDPM. In the stochastic unit commitment case, CLDM shows the lowest dispatch cost, load shedding, and wind curtailment.

In future research, there are two important directions to consider. Firstly, it is crucial to expand the various applications of diffusion models in controllable scenario generation. This will allow for a broader understanding of the potential uses and benefits of diffusion models in scenario generation. Secondly, it is necessary to delve deeper into developing diffusion models specifically tailored for renewable energy scenario generation. This will contribute to enhancing the interpretability and modeling accuracy, thereby improving their practical utility.

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